Malo Leroy

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Experience

Research Scientist Intern | IBM Quantum 2025 - 2026

Theoretical and experimental research in Quantum Error Mitigation (QEM) **Qiskit, SciPy, NumPy**

- Conducted research on a novel QEM technique in a mixed approach of quantum information, data science and algorithmics
- Collaborated with IBM researchers to implement and benchmark the technique on real quantum hardware

Exploratory research on quantum circuit synthesis for dynamical quantum algorithms **Python, QASM**

- Designed a novel data structure based on decision diagrams to simulate and synthetize dynamical algorithms
- Built a toy implementation to test show applications to surface code simulation and broader Quantum Error Correction (QEC)

Software Engineer | Paris Digital Lab 2025

Highly selective gap year program developing three 7-weeks innovative projects for companies **Rust, C++, Flutter, Docker**

- Programmed a peformance-focused, Cloud-native Fully Homomorphic Encryption framework for BPCE Group
- Engineered a secure phone dialer and messaging app with identity verification based on the european eIDAS / EUDI standard

Quantum Computing Research Intern | INRIA 2023 - 2025

Part-time intern in INRIA's QuaCS team **QASM, C++, GoogleTest, CMake**

- Theorized an innovative data structure to represent quantum states & circuits
- Started writing a scientific paper on the subject, detailing the mathematical background and properties of the data structure
- Created and implemented algorithms for fast classical simulation of quantum circuits

General Secretary & Head of Training | ViaRézo (IT association) 2023 - 2025

Technical administrator of France's largest student ISP **Kubernetes, Ansible, OpenStack, Ceph**

- Operated 400+ Unifi APs and 50+ Juniper switches to provide Internet to 2.5k+ students
- Held legal responsibility for database administration (13M+ rows), ensuring european and national compliance (GDPR, NIS2)
- Devised and conducted public training sessions and internal workshops on various technical topics (Bash, Docker, LaTeX, Git)

Network Infrastructure Intern | Paris-Saclay University 2024

- Troubleshooted and secured a university library's network infrastructure **Cisco IOS, Wireshark, VLANs**

Projects

Quantum Semantics for Natural Language Processing | Academic project 2026

- Designed a trainable parameterized quantum circuit for a GPT-like embedding and text-generation model **Python, Qiskit**
- Evaluated the model against a classical baseline and against quantum embedding approaches requiring more context

C/C++ Unit testing library | Personal project 2022

- Developed a fast and memory-safe library leveraging the preprocessor, with a stack-based approach **C, GitHub Actions**
- Engineered pipelines for continuous integration and live documentation deployment

Classical simulation of quantum algorithms | Chateaubriand Highschool 2022

- Created a formal algebra library to enable exact scalar and matrix computations **Python**
- Programmed a standalone proof-of-concept implementation of Shor's algorithm on up to 10 qubits

Education

CentraleSupélec – Paris-Saclay University 2023-2027

Master of Engineering (MEng), current GPA 3.97

- CentraleSupelec is an elite French engineering school part of Paris-Saclay University (12th in the Shanghai Ranking)
- **Relevant Coursework:** quantum compilation, information theory, quantum physics, microarchitecture design & assembly, linear & general algebra, design and verification of formal systems

QMI master's program – Institut Polytechnique de Paris 2026-2027

Master of Science (MSc), no grades yet

- Focused on theoretical computer science for quantum, in addition to the CentraleSupélec curriculum, as a dual-degree
- **Relevant Coursework:** tensor networks, quantum hardware, quantum information theory, quantum non-locality, quantum error correction, quantum structures, ZX calculus, quantum compilation, quantum machine learning

Chateaubriand Highschool 2021-2023

Selective post-secondary school

- Intensive two-year preparatory program for competitive entrance exams to French Grandes Ecoles, with a focus on mathematics

Languages: French (native), English (fluent), Spanish (conversational), German (basic)